# A.1. Linear algebra

*Linear algebra* is the study of linear transformations. A *linear transformation* is a transformation (e.g., a function) in which the sum of the transformation of two inputs separately, such as $T(a)$ and $T(b)$, is the same as summing the two inputs and transforming them together, i.e., $T(a + b) = T(a) + T(b)$. A linear transformation also has the property that $T(a \times b) = a \times T(b)$. Linear transformations are said to preserve the operations of addition and multiplication since you can apply these operations either before or after the linear transformation and the result is the same.

One informal way to think of this is that linear transformations do not have "economies of scale." For example, think of a linear transformation as converting money as the input into some other resource, like gold, so that $T(\$100) = 1$ *unit of gold*. The unit price of gold will be constant no matter how much money you put in. In contrast, nonlinear transformations might give you a "bulk discount," so that if you buy 1,000 units of gold or more, the price would be less on a per unit basis than if you bought less than 1,000 units.

Another way to think of linear transformations is to make a connection to calculus (which we'll review in more detail shortly). A function or transformation takes some input value, $x$, and maps it to some output value, $y$. A particular output $y$ may be a larger or smaller value than the input $x$, or more generally a *neighborhood* around an input $x$ will be mapped to a larger or smaller neighborhood around the output $y$. Here a *neighborhood* refers to the set of points arbitrarily close to $x$ or $y$. For a single-variable function like $f(x) = 2x + 1$, a neighborhood is actually an interval. For example, the neighborhood around an input point $x = 2$ would be all the points arbitrarily close to 2, such as 2.000001 and 1.99999999.

One way to think of the derivative of a function at a point is as the ratio of the size of the output interval around that point to the size of the input interval around the input point. Linear transformations will always have